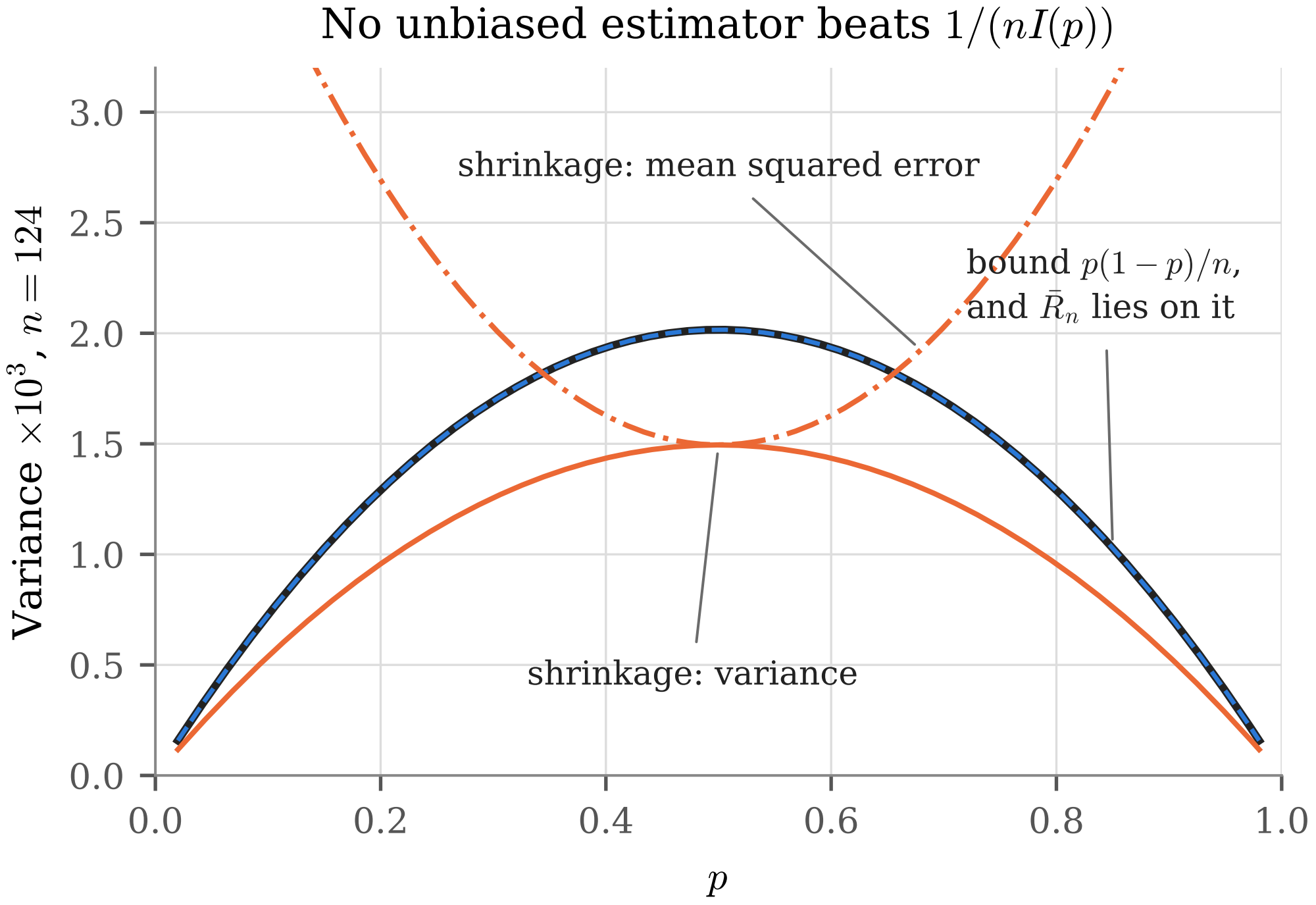
No unbiased estimator beats $1/(nI(p))$
shrinkage: mean squared error
bound $p(1-p)/n$,
and $\bar{R}_n$ lies on it
shrinkage: variance
Variance $\times 10^3$, $n = 124$
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
p